International Journal of Heat and Mass Transfer 164 (2021) 120403

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/hmt

Assessment of nanofluids pool boiling critical heat flux

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ARTICLE INFO

Article history:
Received 22 May 2020
Revised 5 August 2020
Accepted 26 August 2020
Available online 8 October 2020

Keywords:
Pool boiling
Nanofluid
Critical heat flux (CHF)
Contact angle
Capillary wicking

ABSTRACT

Despite contradiction in nucleate pool boiling heat transfer coefficient of nanofluids, the enhancement in critical heat flux (CHF) has been widely approved. This study is dedicated to the assessment of pool boiling CHF of the nanofluids. The assessment is based on a new consolidated CHF database consisting of 431 data points amassed from 62 sources, and includes 14 nanoparticle materials and 4 base fluids, pressures from 3 to 1100 kPa, orientation angles from 0 to 150°, and covering a variety of particle concentrations. It is shown that the maximum CHF ratio of nanofluid to base fluid can achieve 4 and 4.6 for upward-facing heater and inclined heater, respectively. The heater type (wire or block) has an important effect on CHF, as the direct electric-heating wire heater causes electrophoresis, accelerating nanoparticle deposition. The CHF ratio is reduced with increasing pressure in the present range, but nanoparticle size has a minor effect on it, which is more determined by the dispersion level of particles in base fluid. Two major mechanisms, including improved wettability and enhanced capillary wicking caused by nanoparticle deposition, are responsible for the enhancement of CHF, latter of which warrants more theoretical studies in the future.

1. Introduction

1.1. Background and applications

The recent advancements in applications such as high-performance computers, electrical vehicle power electronics, avionics, in-vessel retention (IVR), and directed energy laser and microwave weapon systems, have led to unprecedented increases in power density, rendering obsolete all fan-cooled heat sinks and even some of the most aggressive single-phase liquid cooling schemes. These increases have necessitated a transition to two-phase cooling schemes, which capitalize upon the coolant's both latent and sensible heat content to achieve orders of magnitude enhancement in boiling and condensation heat transfer coefficients. Despite a mass of studies for understanding of several advanced two-phase cooling schemes in recent years, such as micro-channel, spray cooling and jet impingement, pool boiling is still the most prevalent and reliable cooling method due to its simplicity in both construction and implementation in practical applications.

1.2. Boiling curve and necessities for nanofluids boiling enhancement

Heat transfer response in pool boiling can be described by the boiling curve [1,2], as shown in Fig. 1. The boiling curve depicts the variation of heat flux from the surface to the pool liquid with wall superheat (T_w - T_{sat}). It is highly effective at identifying the different heat transfer regimes encountered at different levels of wall superheat. They are comprised of (a) single-phase regime corresponding to low superheats, (b) nucleate boiling regime associated with bubble nucleation within liquid and highest heat transfer coefficients, (c) transition boiling regime associated with portions of the surface encounter bubble nucleation while other regions are blanketed with vapor, and (d) film boiling regime corresponding to high wall superheats that cause vapor blanketing over the entire surface and very small heat transfer coefficients. These four regimes are demarcated by three important transition points: (a) onset of boiling (ONB) or incipient boiling corresponding to first bubble formation on the surface, (b) critical heat flux (CHF), where bubble nucleation in nucleate boiling is replaced by localized vapor blankets merging together across the surface, and (c) minimum heat flux (or Leidenfrost point), below which partial breakup of the continuous vapor blanket associated with film boiling begins to take place. The three transition points clearly have profound influence on pool boiling heat transfer effectiveness.

The present study is focused entirely on the CHF point, which marks the transition from the much fast and steady nucleate

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https://doi.org/10.1016/j.ijheatmasstransfer.2020.120403
0017-9310/

Nomenclature

g	gravitational acceleration
h_{fg}	latent heat of vaporization
P	pressure
q''	heat flux
T_{sat}	saturation temperature
T_w	wall temperature
Greek symbols	
α	contact angle
θ	heater orientation angle
ρ	density
σ	surface tension
Subscripts	
bf	base fluid
CHF	critical heat flux
f	liquid
g	vapor
max	maximum
nf	nanofluid

boiling to the deficient and transient transition boiling. CHF is also termed as 'boiling crisis', represented by a sharp increase of heater surface temperature due to the dryout of heater surface, accompanied by the ultimate physical destruction or burnout of the heater. Thus, optimum cooling will be achieved by maintaining conditions above onset of boiling, but safely below CHF, rendering the CHF point as the most important design and safety parameter in pool boiling. However, the absence of pumping devices in pool boiling leads to weak liquid motion, as a consequence the heat transfer efficiency is weakened. And, conventional heat transfer liquids, including water, refrigerant, oil, and ethylene glycol, whose properties are relatively poor, resulting in significant limits for heat transfer performance, which has become a major obstacle not only for the development of heat exchange devices with high compactness and effectiveness, but also for the use of pool boiling in high-heat-flux cooling applications. As the inclusion of pumping facilities is costly and inconvenient under compact situations, other

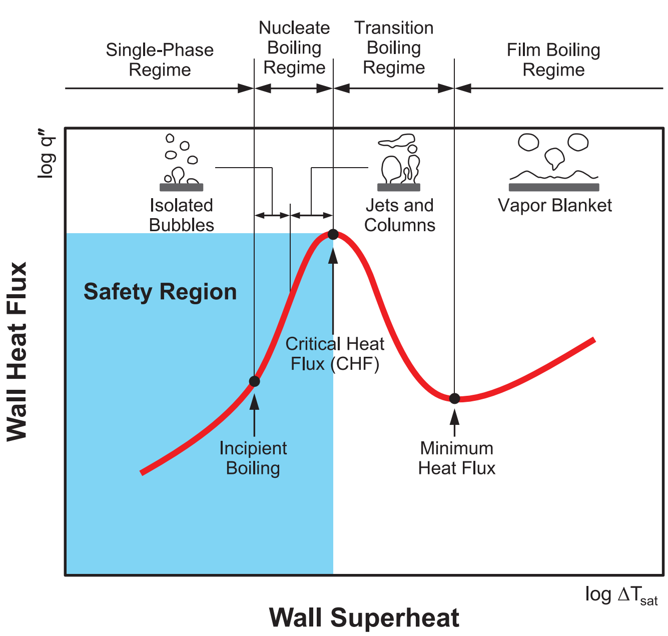

Fig. 1. Boiling curve.

techniques by modifying liquid properties will be a good option for enhancing boiling heat transfer performance, and nanofluid is one of these, which attracted significant attention in the past two decades.

Nanofluid is a type of fluid containing nanoparticles (typically 1–100 nm in size) that are uniformly and stably suspended in a liquid [3], as shown in Fig. 2(a). The concept of nanofluid was first proposed by Choi and Eastman [4] in 1995, despite that the potential for ultra-fine 13-nm particles dispersed in fluids in altering fluid thermal conductivity and viscosity has been observed by Masuda et al. [5] before 1995. Since then, the nanofluid has attracted considerable attention and a mass of studies can be found for nanofluid single-phase heat transfer. In general, nanoparticle materials include chemically stable metals, metal oxides and carbon in various forms [6]. Nanofluid pool boiling was not reported until the early 2000s, by Das et al. [7] and You et al. [8] in 2003. Particularly, the capability in improving CHF by using nanofluids was brought to public attention by the pioneering work of You et al., who reported a two-fold CHF enhancement by using water-based alumina nanofluids. Because of its much great potential in CHF enhancement, researchers in the heat transfer community conducted numerous studies in pursuit of better understanding of nanofluids CHF in the past decade.

1.3. Practical concerns and potential in in-vessel retention (IVR)

Despite contradiction in the boiling heat transfer coefficient of nanofluids, caused by complexities by the presence of suspended nanoparticles with a variety of particle types, sizes, preparing methods, functionalization methods and concentrations, it has been widely accepted that nanofluids can enhance pool boiling CHF significantly [10], schematized in Fig. 2(b). On the other hand, there are several practical concerns related to the use of nanofluids in boiling facilities as reported in a previous review [10], like clustering, sedimentation and precipitation of nanoparticles, clogging of flow passages, erosion to heating surface, nanoparticle deposition, transient heat transfer performance, high cost of nanofluids and difficulties in production, and lack of quality assurance, the nanofluids can be potentially used where only CHF is concerned. For example, nanofluids can be implemented to improve the capability of IVR in the severe accident management strategy in a light-water reactor, which are stored in a separate system at the normal operating conditions, but discharge into the reactor cavity through injection lines within a few seconds upon actuation. As assessed by Buongiorno et al. [11,12], the enhancement in the decay power removal through the vessel using nanofluids is about 40%, which provides a higher IVR safety margin or, for a given margin, to enable IVR at higher core power. Their studies prove the superiority of nanofluids over pure liquid in IVR. Therefore, there is a need to fully understand the data trends, enhancement magnitudes, and mechanisms of nanofluids pool boiling CHF with different nanoparticle materials and base fluids.

1.4. Objective of study

The present study will build up a consolidated database of nanofluids pool boiling CHF, amassed from 62 sources, which consists of 431 data points, and includes variations in particle material and size, base fluid, heater geometrical shape and orientation, pressure, and contact angle. Using this consolidated database, the maximum enhancements in CHF with different nanoparticle materials and base fluids are assessed at first. Then the effects of heater type and orientation, operation pressure and particle size on CHF are clarified. Finally, the mechanisms of CHF are discussed concerning wettability and capillary wicking induced by the

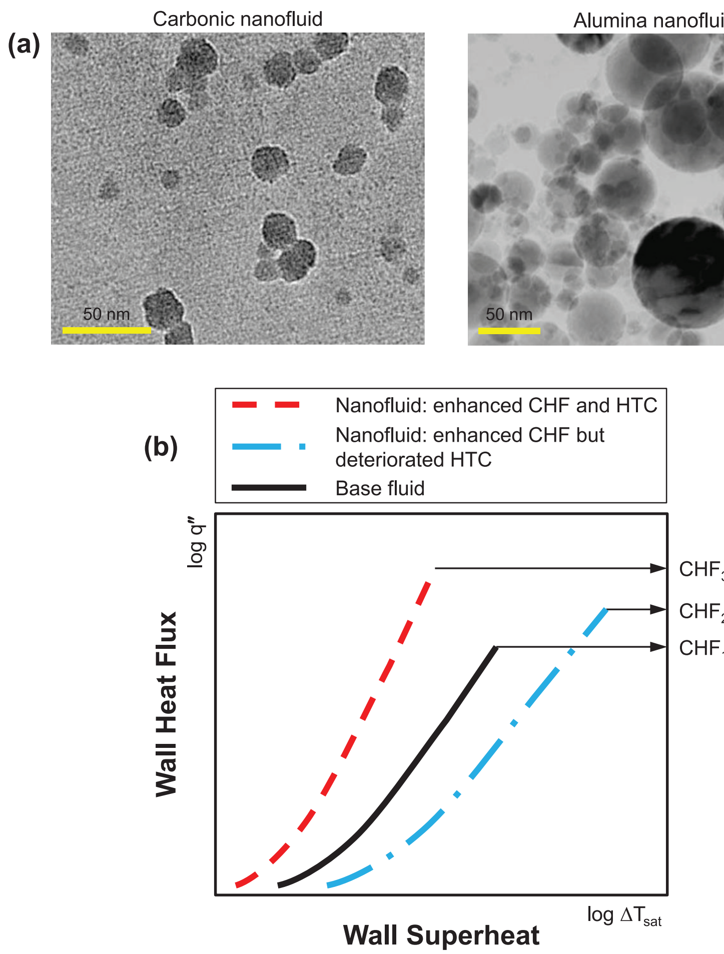

Fig. 2. (a) TEM images of water-based carbonic nanofluid by one-step method and alumina nanofluid by two-step method, adapted from Kim et al. [9]. (b) Comparison of boiling curves between nanofluids and base fluid.

deposition of nanoparticles on the surface. Based on the assessment, better understanding of nanofluids pool boiling CHF will be facilitated.

2. New consolidated database for nanofluids CHF

In the present study, a total of 431 data points of saturated pool boiling CHF for 14 kinds of nanoparticle materials, including Al_2O_3, SiO_2, TiO_2, carbon nanotubes (CNT), graphene, Fe_3O_4, CuO, SiC, ZnO, diamond, Cu, BiO_2, ZrO_2 and Fe_2O_3, and 4 kinds of base fluids, including water, ethanol, LiBr-water mixture, and ethylene glycol-water mixture are consolidated. These data are amassed from 62 sources in the public literatures, obtained either directly from the original sources or extracted from digitalized figures using commercial software. Notice that all the CHF data included in the database are under steady state and saturated conditions. Data from transient heat transfer or quenching of nanofluids, and nanolubricant in refrigeration are excluded. Table 1 provides key information on the individual databases incorporated in the consolidated database. Despite the popularity of composite nanofluids (two or more types of nanoparticles dispersed in the base fluid) in nano community [13], CHF for those composite nanofluids is quite sparse at present. Recently, Reddy and Venkatachalapathy [14] reported that the composite Al_2O_3 and CuO nanofluid showed higher CHF than the single type of nanofluid. However, superiority of the composite nanofluid in boiling needs to be verified further.

Fig. 3 shows the source percentage for each nanoparticle material in the consolidated database. It is clear that Al_2O_3 and SiO_2 are the most common materials used in nanofluids pool boiling experiments, followed by TiO_2 and CNT. As will be discussed later, the heat transfer augmentation using nanofluids can be attributed to the nanoparticle deposition, which is a major practical concern for the deployment of nanofluids in boiling applications. Fe_3O_4 nanofluids, however, warrant more studies in the future because the nanoparticle deposition can be well controlled by configuring an external magnetic field in the cooling loop. There is no doubt that water is the most common base fluid adopted, even though it is not demonstrated in the figure. Three types of unit representing nanoparticle concentration are often used: volume fraction (vol.%), weight fraction (wt.%) and gram per liter (g/L).

Overall, the consolidated database consists of 431 nanofluids pool boiling CHF data points from 62 sources with the following coverage:

- Nanoparticles: Al_2O_3, SiO_2, TiO_2, carbon nanotubes (CNT), Fe_3O_4, graphene, CuO, SiC, ZnO, diamond, Cu, BiO_2, ZrO_2 and Fe_2O_3.
  Base fluids: water, ethanol, LiBr-water mixture and ethylene glycol-water mixture with various fractions.
  Particle concentration: 0–4 vol.%, 0–5.25 wt.%, 0–10.07 g/L.
  Pressure: $3 \le P \le 1100$ kPa.
  Orientation angle: $0 \le \theta \le 150°$.

Table 1
CHF data for nanofluids included in the consolidated database.

Author(s)	Pressure	Heater material	Heater size width × length or diameter (thickness) [mm]	Base fluid	Particle material	Particle size [nm]	Particle concentration	Number of data points
Bang and Chang [15]	atmospheric	copper	4 × 100 (1.9)	water	Al_2O_3	47	0–4 vol.%	8
Kim et al. [6]	atmospheric	stainless steel	0.381	water	Al_2O_3, SiO_2, ZrO_2	20–250	0.001–0.1 vol.%	9
Bang et al. [16]	atmospheric	sapphire	24 × 10 (1)	ethanol	Al_2O_3	118.2	0.01 vol.%	1
Park et al. [17]	atmospheric	NiCr	0.98	water	Al_2O_3, graphene, graphene-oxide	$<$ 45 μm	0.001 vol.%	3
Lee et al. [18]	atmospheric	NiCr	0.49	water	CuO	15	0.001 vol.%	2
Park et al. [19]	atmospheric	NiCr	0.49	water	graphene-oxide, Al_2O_3, SiO_2	$<$ 45 μm	0.0001 vol.%	9
Park and Bang [20]	atmospheric	NiCr	0.49	water	ZnO, SiC, CuO, graphene-oxide, Al_2O_3, SiO_2	$<$ 45 μm	0.01 vol.%	6
Song et al. [21]	atmospheric	stainless steel	10 × 50 (0.4), 50 × 50 (0.4)	water	SiC	$<$ 100	0.0001–0.01 vol.%	9
Gerardi et al. [22]	atmospheric	ITO	10 × 30 (0.0007)	water	diamond, SiO_2	$\leq$ 173	0.01–0.1 vol.%	5
Ahn et al. [23]	atmospheric	SiO_2	20 × 25	water	reduced graphene-oxide	0.5–1.0 μm	0.0001–0.001 wt.%	7
Kim et al. [24]	atmospheric	NiCr, Ti	0.2–0.25	water	Al_2O_3, TiO_2	47–85	0.00001–0.1 vol.%	13
Kim and Kim [25]	atmospheric	NiCr	0.2	water	SiO_2	90	0.0001–1 vol.%	5
Kim et al. [26]	atmospheric	copper, nickel	20	water	Al_2O_3, TiO_2	45–47	0.01 vol.%	6
Jung et al. [27]	atmospheric	copper	20	water	TiO_2	27.2–85.3	0.00001–1 vol.%	12
Jung et al. [28]	T_{sat} = 29 °C	copper	10 × 10 (30)	water	Al_2O_3	45	0.00001–0.1 vol.%	10
Jung et al. [29]	4 kPa	copper	10 × 10 (30)	LiBr-water mixture	Al_2O_3	46	0.01–0.1 vol.%	12
Kathiravan et al. [30]	atmospheric	stainless steel	10.6	water	Cu	10–20	0.028–0.113 vol.%	4
Kathiravan et al. [31]	atmospheric	stainless steel	30	water	Cu	10–20	0.028–0.113 vol.%	6
Kathiravan et al. [32]	atmospheric	stainless steel	30	water	CNT	10–15	0.25–1 vol.%	6
Vazquez and Kumar [33]	atmospheric	NiCr	0.79 × 0.127	water	SiO_2	10	0.1–2 vol.%	21
He et al. [34]	atmospheric	NiCr	0.5	ethylene glycol-water mixture	ZnO	30	5.25 wt.%	3
Hu et al. [35]	atmospheric	platinum	0.07	ethylene glycol-water mixture	graphene	0.335–3.35	0.005–0.1 wt.%	5
Liu et al. [36]	7.4–103 kPa	copper	40 × 40	water	CNT	15	0.5–2.5 wt.%	15
Yang and Liu [37]	7.4–103 kPa	copper	40 × 40	water	SiO_2	30	1.5 wt.%	3
Coursey and Kim [38]	T_{sat} = 29.8 °C	copper	2 cm^2	water, ethanol	Al_2O_3	45	0.026–10.07 g/L	11
Golubovic et al. [39]	atmospheric	NiCr	0.64	water	Al_2O_3, BiO_2	22.6–46	0.0005–0.01 vol.%	20
Hegde et al. [40]	atmospheric	NiCr	0.19	water	Al_2O_3	10–80	0.01–0.09 vol.%	5
Hegde et al. [41]	atmospheric	NiCr	0.19	water	CuO	50	0.01–0.5 vol.%	7
Lee et al. [42]	atmospheric	NiCr	0.4	water	Fe_3O_4, Al_2O_3, TiO_2	15–80	0.0001–0.01 vol.%	11

(*continued on next page*)

Table 1 (*continued*)

Author(s)	Pressure	Heater material	Heater size width × length or diameter (thickness) [mm]	Base fluid	Particle material	Particle size [nm]	Particle concentration	Number of data points
Lee et al. [43]	101–1100 kPa	NiCr	0.4	water	Fe_3O_4, Al_2O_3	20–30	0.0001 vol.%	5
Sheikhbahai et al. [44]	atmospheric	NiCr	____	ethylene glycol-water mixture	Fe_3O_4	< 50	0.01–0.1 vol.%	3
Ahn and Kim [45]	atmospheric	copper	10	water	Al_2O_3	47	0.001 vol.%	1
Kim et al. [46]	4 kPa	NiCr	0.2	water	Al_2O_3	40–50	0.00001–0.1 vol.%	5
Mourgues et al. [47]	atmospheric	stainless steel	15	water	ZnO	____	0.01 vol.%	2
Shahmoradi et al. [48]	atmospheric	copper	38	water	Al_2O_3	40	0.001–0.1 vol.%	5
Amiri et al. [49]	atmospheric	stainless steel	15	water	CNT	< 60	0.01–0.1 wt.%	13
Kole and Dey [50]	atmospheric	Constantan wire	0.07	water	Cu	40–60	0.005–0.5 wt.%	3
Lee et al. [51]	atmospheric	NiCr	0.4	water	Fe_3O_4	25	0.0001–0.1 vol.%	6
Park and Kim [52]	19.61 kPa	zirconium	9.53 × 9.53 (4)	water	CNT	10–15	0.0001–0.1 vol.%	12
Kamatchi et al. [53]	atmospheric	NiCr	0.45	water	reduced graphene-oxide	2.98	0.01–0.3 g/L	3
Sakashita [54]	0.1–0.8 MPa	copper	7	water	TiO_2	25	0.002 wt.%	6
Ciloglu [55]	atmospheric	copper	40	water	SiO_2	20	0.01–0.1 vol.%	3
Dadjoo et al. [56]	atmospheric	copper	30	water	SiO_2	7–14	0.001–0.005 vol.%	9
Ham et al. [57]	3 kPa	copper	____	water	Al_2O_3	50	0.001–0.1 vol.%	8
Hu et al. [58]	____	platinum	0.07	ethylene glycol-water mixture	SiO_2	120	0.25 vol.%	1
Sulaiman et al. [59]	atmospheric	copper	20	water	SiO_2, Al_2O_3, TiO_2	13–21	0.04–1 g/L	18
Vassallo et al. [60]	atmospheric	NiCr	0.4	water	SiO_2	15–50	0.5 vol.%	2
Park et al. [61]	T_{sat} = 60 °C	copper	9.53 × 9.53 (4)	water	CNT	10–20	0.0001–0.05 vol.%	4
Park et al. [62]	T_{sat} = 60 °C	copper, titanium	9.53 × 9.53 (4), 0.25	water	CNT	10–20	0.0001–0.05 vol.%	7
Sarafraz et al. [63]	____	copper	11	ethylene glycol-water mixture	ZrO_2	20–25	0.025–0.1 wt.%	4
Salari et al. [64]	____	copper	11	water	Fe_3O_4	20	0.1–0.3 wt.%	3
Sarafraz et al. [65]	atmospheric	copper	11	water	CNT	10–20	0.1–0.3 wt.%	3
Salari et al. [66]	atmospheric	copper	11	water	TiO_2	40–50	0.1–0.3 wt.%	3
You et al. [8]	T_{sat} = 60 °C	copper	10 × 10	water	Al_2O_3	____	0.001–0.05 g/L	5
Kim et al. [67]	T_{sat} = 60 °C	copper	10 × 10	water	Al_2O_3	32	0.025 g/L	5
Kwark et al. [68]	atmospheric	copper	10 × 10 (3)	water	Al_2O_3, CuO, diamond	86–139	0.001–1 g/L	21
Rostamian and Etesami [69]	atmospheric	copper	28	water	SiO_2	7–14	0.005 vol.%	1
Dareh et al. [70]	atmospheric	copper	20 × 20 (5)	water	Al_2O_3	20–25	0.0025–0.01 vol.%	3
Kumar et al. [71]	atmospheric	NiCr	0.27	water	Fe_2O_3, Al_2O_3, CuO	10–26	0.01–0.1 vol.%	15
Hu et al. [72]	atmospheric	platinum	0.07	water	SiO_2	98	0.081–0.325 wt.%	3
Yao et al. [73]	70–130 kPa	copper	____	water	Al_2O_3, SiO_2	30–50	0.01 wt.%	10
Kshirsagar and Shrivastava [74]	atmospheric	NiCr	____	water	Al_2O_3	30	0.3–1.5 wt.%	5

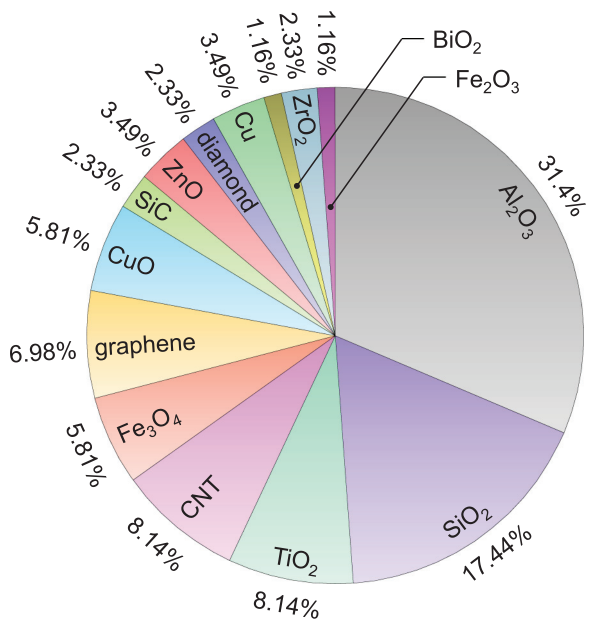

Fig. 3. Source percentage for nanoparticle material in present consolidated database.

3. Assessment of nanofluids CHF

3.1. Maximum enhancement for different particle materials and base fluids

Shown in Fig. 4 is the ratio of maximum nanofluid pool boiling CHF to base fluid CHF for each nanoparticle material included in the present consolidated database, in which the heater is orientated upward-facing and the base fluid is water. It shows that an extra high CHF ratio of 4 is achieved, reported by Vazquez and Kumar [33] using SiO_2 nanofluids. Namely, the CHF for nanofluids is enhanced by 3 folds. This is followed by the Al_2O_3 nanofluids in Kshirsagar and Shrivastava [74] and the CNT nanofluids in Liu et al. [36], the CHF ratios for both of which are over 3. Notice that Fig. 4 only presents the maximum ratio for each nanoparticle material that the authors have explored in the literatures. It does not mean that SiO_2, Al_2O_3 and CNT nanofluids are superior over the rest nanofluids. One major reason for this is that these three materials are the most investigated by different groups (as shown in Fig. 3), which therefore cover a variety of conditions in experiments. In contrast, other nanofluids are less concerned. In fact, just as reported by Sulaiman et al. [59], the CHF enhancement is weakly dependent on particle material. Instead, it is more determined by the fine distribution of particles within the base fluid, and the deposition rate on the heating surface as will be discussed later. Aside from these two major factors, many other effects, such as particles' own wettability [75], stabilizer (ions and surfactant) added or not [27,28,30,31,76], with or without sonication in nanofluids preparation [51], nanofluid storage time [51] and boiling time [77], and the type of stabilizer [61,62], all can affect the extent of CHF improvement. Overall, the nanofluids show their great potential in improving pool boiling CHF. Even the smallest CHF ratio in Fig. 4 is exceeding 1.5 for BiO_2 nanofluids, extracted from only one source of Golubovic et al. [39]. Thus, more attentions and efforts still need to be paid in practical applications where CHF enhancement is concerned.

Shown in Fig. 5 is the ratio of maximum nanofluid pool boiling CHF to base fluid CHF for various base fluids adopted in the literatures, in which the heater orientation is upward-facing. Since a majority of studies used water as the base fluid, it is not presented in this figure. It shows that the overall enhancement ratios for LiBr-water solution, ethylene glycol (EG)-water solution and ethanol are lower than that for water in Fig. 4, all of which are below 2, with the maximum value of 1.95 achieved by Sheikhbahai et al. [44] using 50 vol.% EG in water. This result can be attributed to the superior thermophysical properties of water, especially the latent heat, far higher than the other base fluids. Besides, the ratio also has a strong dependence on the fraction of EG or LiBr in water, and nanoparticle concentration in base fluids. On the other hand, Coursey and Kim [38] found that the wetting system between base fluid and surface can affect the CHF enhancement significantly: the poorly wetting system, like water on polished copper, will be enhanced by the addition of nanoparticles, but the better wetting system, like ethanol on glass, shows no improvement or even degradation. As the electronic cooling have stringent requirements for dielectricity, 3 M company's Fluorinert and Novec fluids, *e.g.*, FC-72 and HFE-7100, are preferable, therefore the nanofluids based on these dielectric fluids should be paid more attention.

3.2. Differences between wire heater and block heater

In this study, the authors note that CHF for the base fluid of pure water varies greatly in different sources. Shown in Fig. 6(a) presents water CHF from 14 different sources for the upward-facing orientation under atmospheric pressure. Notice that Fig. 6(a) shares the same x-coordinate with Fig. 6(b). Also included are the results calculated by the classical Zuber model [78-80],

$$q''_{CHF} = 0.131\rho_g h_{fg}\left[\sigma g\left(\rho_f - \rho_g\right)/\rho_g^2\right]^{1/4}, \quad (1)$$

and the predicting method reformed by Liang and Mudawar [81],

$$q''_{CHF} = 0.151[1 - 0.0012\theta tan(0.414\theta) - 0.122sin(0.318\theta)] \times \rho_g h_{fg}\left[\sigma g\left(\rho_f - \rho_g\right)/\rho_g^2\right]^{1/4}, \quad (2)$$

where θ is orientation angle of the heater. Notice that Eq. (2) is a combination of the interfacial lift-off model proposed by Mudawar et al. [82] and the correlation by Chang and You [83]. This reformed predicting method has been validated using a consolidated CHF database for pure liquids consisting of 800 data points amassed from 37 sources, and including 14 working fluids. Fig. 6(a) shows that the consistency of water CHF in different studies is very poor, and several results obtained by Bang and Chang [15], Kim et al. [26], Ahn and Kim [45], Shahmoradi et al. [48], Kumar et al. [71] and Golubovic et al. [39], deviate significantly from the predicted values. This implies that the CHF ratio of nanofluid to base fluid has a strong dependence on the testing facilities utilized in different research groups, especially the heater immersed in boiling liquid. Other factors, like preparation, material, and size of the substrate, are also responsible for the difference. It is very important for researchers to validate their testing facilities strictly using the base fluid before testing nanofluids boiling, to make their testing results more reliable.

Two types of heater are commonly used in nanofluids pool boiling tests, including NiCr wire heater and copper block heater. The former is heated by electricity, which inevitably leads to electrophoresis in electrolytes, causing nanoparticles in liquid to migrate and accumulate on the heating surface, *i.e.*, accelerating the deposition of nanoparticles. This has been evidenced by Santillán et al. [84] and White et al. [85]. Thus, the direct electric-heating technique using the wire heater might be questionable due to the accelerated nanoparticle deposition that distorts the boiling characteristics of nanofluids [26]. Instead, liquid covering the block heater is heated by thermal conduction, and there is no such additional effects. Fig. 6(b) compares the ratio of maximum nanofluid pool boiling CHF to base fluid CHF for the wire and block heaters,

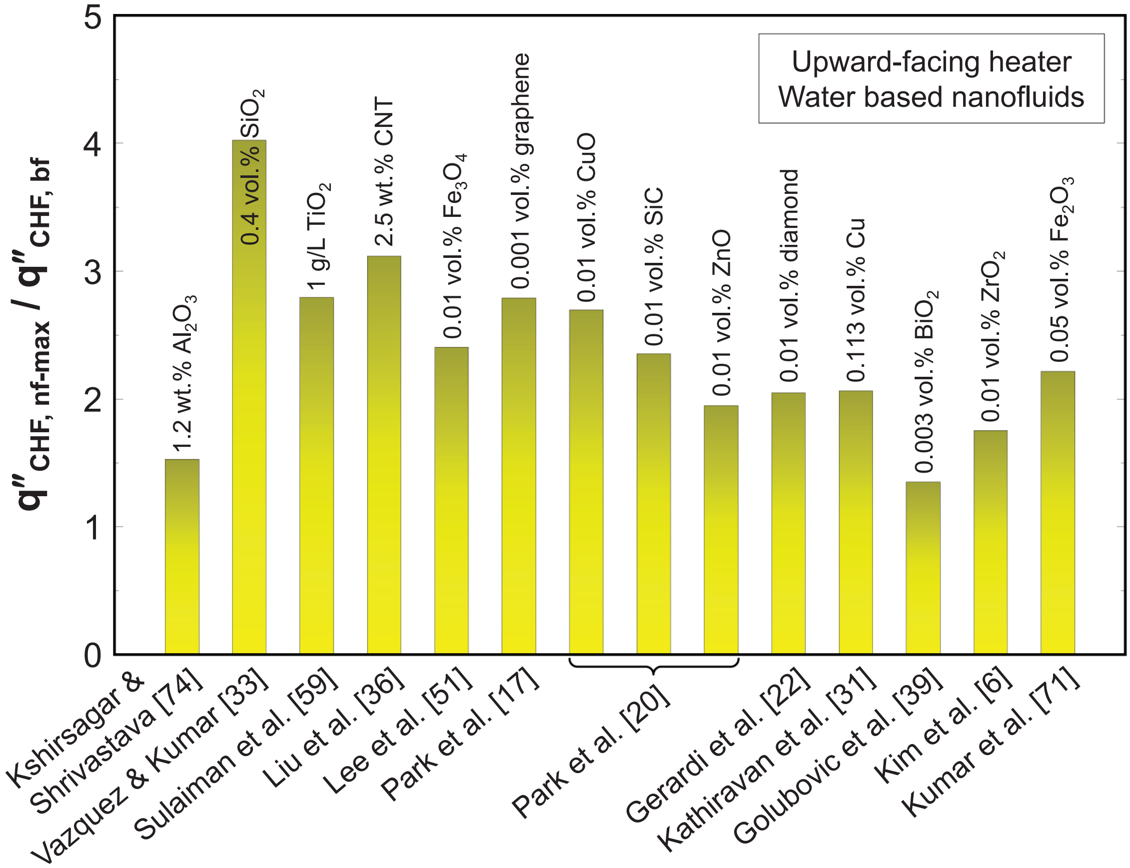

Fig. 4. Maximum CHF ratio of nanofluid to base fluid for various nanoparticle materials.

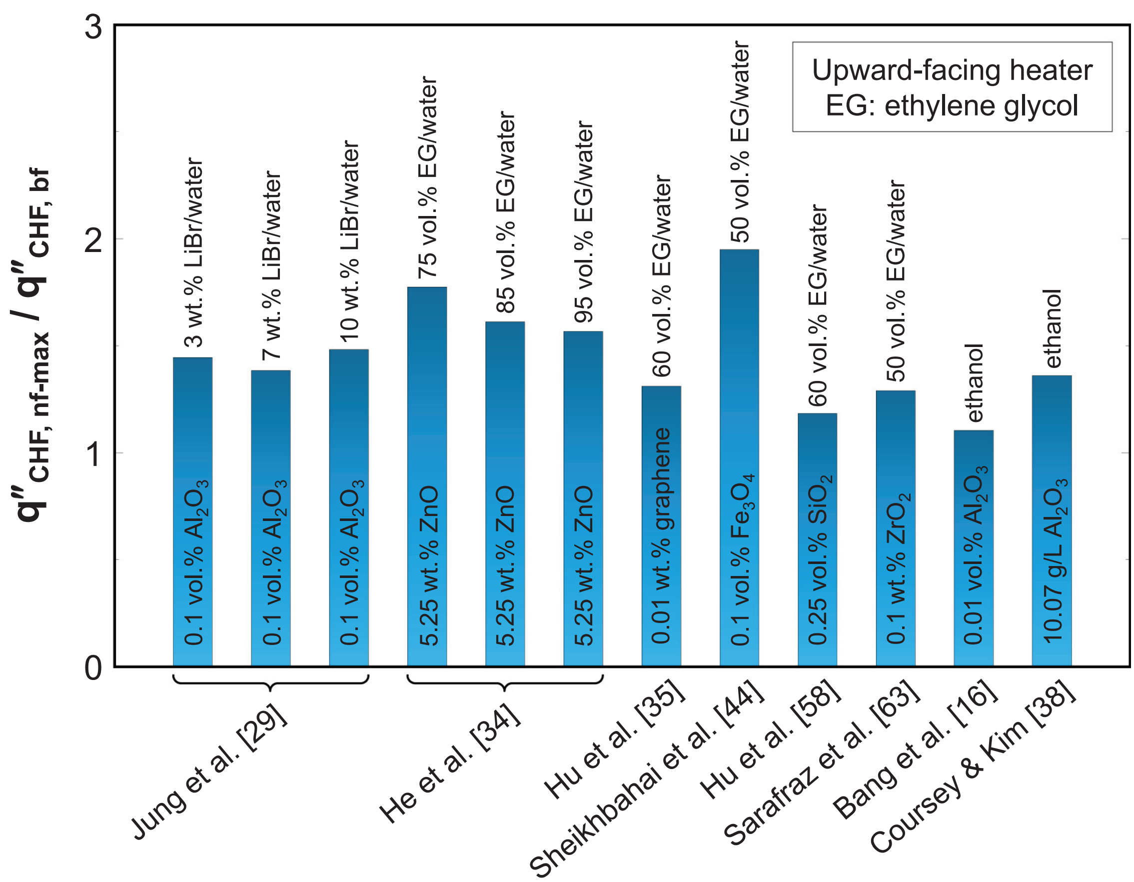

Fig. 5. Maximum CHF ratio of nanofluid to base fluid for various base fluids.

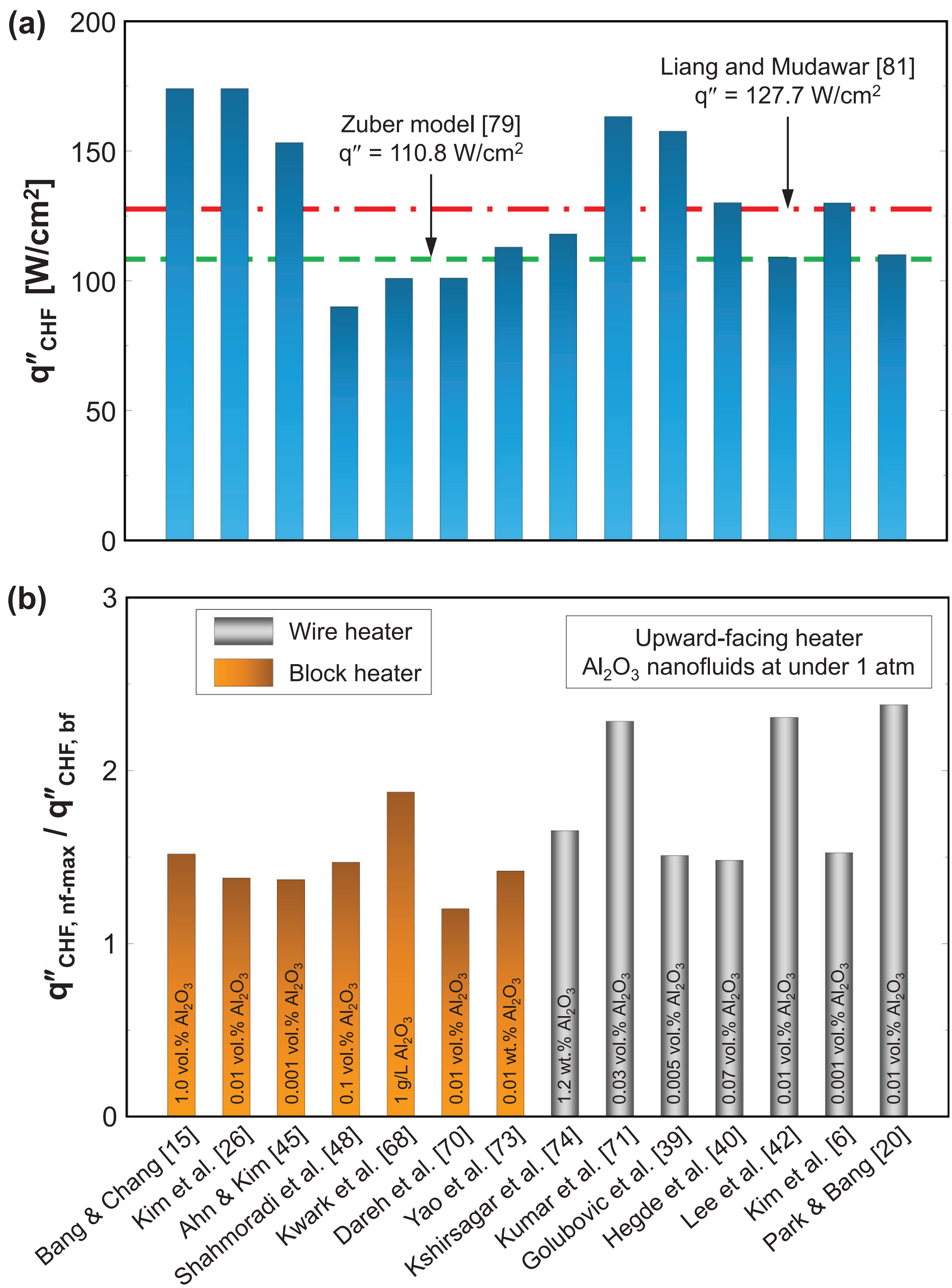

Fig. 6. (a) Water CHF under atmospheric pressure from different sources, and (b) maximum CHF ratio of nanofluid to base fluid for wire heater and block heater.

in which all heaters are oriented upward-facing and using water-based Al_2O_3 nanofluids under atmospheric pressure. It is shown that the CHF ratios using wire heaters are fairly higher than those using block heaters. Aside from the effects of electrophoresis and heater material, size of the heater can also influence pool boiling CHF [86], which however, is not included in the comparison between different studies in Fig. 6(b).

3.3. Parametric effects: pressure, orientation and nanoparticle size

As the cooling process may take place in a confined space, the pressure is not always atmospheric. For example, in severe nuclear reactor accidents, such as reactor core meltdown and core disassembly, the pressure in pressure vessel is extremely high. Thus, there is a need to understand the pressure effect on nanofluids

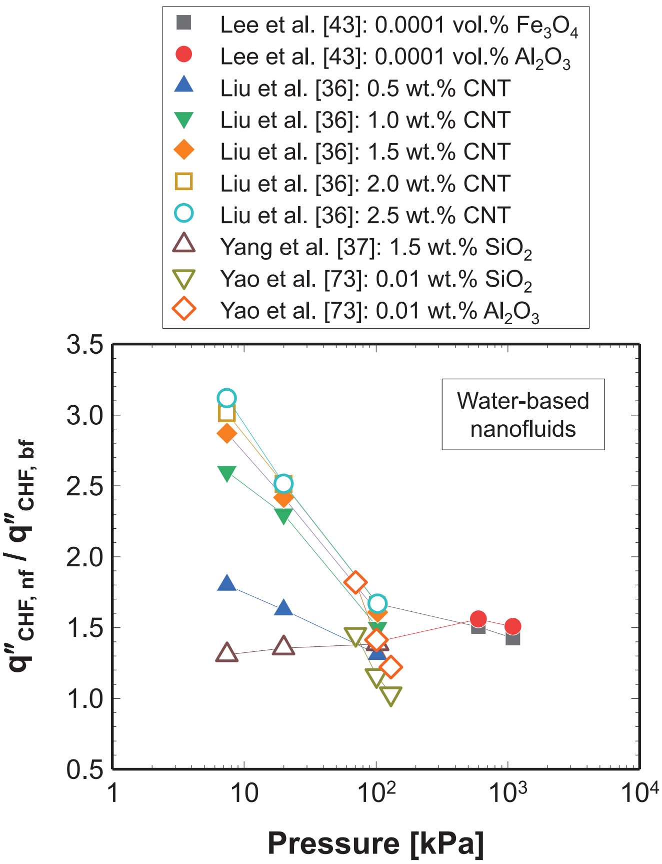

Fig. 7. Effect of operation pressure on CHF ratio of nanofluid to base fluid.

pool boiling CHF. Here the experimental data from Lee et al. [43], Liu et al. [36], Yang et al. [37], and Yao et al. [73] are used to explore the pressure effect, including water-based Fe_3O_4, Al_2O_3, CNT, and SiO_2 nanofluids. Those data are plotted in Fig. 7. Besides the weak dependence of the ratio of nanofluids CHF to base fluid CHF on pressure for the 0.0001 vol.% Al_2O_3 nanofluid and the 0.01 wt.% SiO_2 nanofluid, other CHF ratios decrease with increasing pressure. According to the assessment for pure liquids in Liang and Mudawar [81], CHF for water increases with increasing pressure for the pressure range in Fig. 7. In fact, the increasing trend of nanofluids CHF have also been witnessed in the literatures. This indicates that the positive effect raised by increasing pressure on CHF is weaker for nanofluids than the base fluid. Notice that the maximum pressure in Fig. 7 is 1100 kPa, far smaller than the pressure in pressure vessel used in the light-water reactor. Even for a pure liquid, the variation trend of CHF with pressure is similar to a parabola. As the range of pressure in the existing data is quite narrow, the pressure's effect covering a wide range, better to be near the critical pressure, on the nanofluids CHF enhancement warrants more studies.

Also, several studies addressing the effect of heater orientation on nanofluid pool boiling CHF can be found in the literatures, including Park et al. [19], Kim et al. [67] and Dadjoo et al. [56]. The nanofluids include water-based Al_2O_3, graphene-oxide, and SiO_2 nanofluids. In this study, the horizontal upward-facing orientation is set as $\theta = 0°$, vertical orientation as $\theta = 90°$ and downward-facing as $\theta = 180°$. The data extracted from references [19,56,67] are plotted in Fig. 8(a). Also CHFs of nanofluids and base fluid of water are plotted in Figs. 8(b) and (c), respectively, to assist the understanding of Fig. 8(a). Notice that Figs. 8(a) and (b) share the same set of legends. In Dadjoo et al., the CHF ratio of nanofluid to base fluid increases slightly with increasing the orientation angle, which contradicts with the CHF ratio trend in Park et al. In fact, even for the pure liquids, CHF also decreases with increasing the orientation angle [87-89]. This decreasing trend of the nanofluids CHF ratio with increasing the angle has a weak dependence on the nanoparticle materials of Al_2O_3, graphene-oxide and SiO_2. Aside from the suppression of bubble release from the heating surface with increasing the angle, mainly caused by bubble coalescence and sweeping along the surface because of buoyancy effect, weakening of nanoparticle deposition due to gravity is another reason being responsible for this trend. However, Dadjoo et al. also attributed their increasing trend of CHF with orientation angle to the promoted deposition of nanoparticles, which conflicts with the effect of buoyancy nature on the loose transient nanoparticle layer. Kim et al. used 'g/L' as the unit of nanoparticle concentration rather than vol.% and wt.%. They also showed that the CHF ratio decreases when the orientation angle increases from 0° to 90°. However, for the downward orientation with angles varying in 90°−180°, the CHF ratio increases significantly with increasing the angle, and the highest ratio of 4.6 is achieved at $\theta = 150°$. This implies that the enhancement in CHF is more effective at or near the downward-facing region. There is no doubt that CHFs for both pure liquid and nanofluid decrease when the surface is oriented from vertical to downward-facing because of bubble aggregation under the surface. The increasing of CHF ratio means that the decreasing of nanofluids CHF is less than that of base fluid. The deposition of nanoparticles will create a nanoporous structure on the heating surface, which can induce capillary wicking, assisting surface wetting and alleviating bubble aggregation there due to condensing effect. However, to the authors' knowledge, no more CHF data for nanofluids can be found for the orientation angle in 90°−180°, which points to the need for more experiments and improved understanding of near-wall interfacial behavior for these orientations.

In general, the size of nanoparticles is usually heterogeneous because of the difficulties in manufacturing nanoparticles with the same size. This might be the reason why researchers or vendors used the average size to represent the scale of nanoparticles used to prepare nanofluids. Fig. 9 presents the impact of nanoparticle size on the CHF ratio. As a vast number of studies have shown that the CHF enhancement has a strong dependence on nanoparticle concentration, two Al_2O_3 concentrations of 0.0001 vol.% and 0.01 vol.%, and one SiO_2 concentration of 0.01 vol.% are selected from sources of Kim et al. [24,26], Lee et al. [42, 43], Yao et al. [73], Kumar et al. [71], Ciloglu [55] and Kim et al. [6], in which water is used as the base fluid. All the data plotted in Fig. 9 are under atmospheric pressure and the heaters are oriented upward-facing. For a fixed concentration, Fig. 9 shows that the particle size varying in 10–60 nm has a fairly minor effect on the CHF ratio. As discussed in a prior section, the heat transfer enhancement is more determined by the dispersion level of nanoparticles in the base fluid. A fine distribution of nanoparticles makes the nanofluid be more close to a homogeneous fluid, which is favorable to pool boiling heat transfer enhancement. This is often realized by adding surfactant, polymer or acid into the nanofluids to generate repulsive force between adjacent particles.

4. Mechanisms for CHF enhancement

Researchers have shown that the CHF enhancement using nanofluids is mainly caused by the deposition of nanoparticles [24-26,39,47,90,91]. They validated this assumption by comparing nanofluids CHF on a bare heater with base fluid CHF on a nanoparticle-coated heater prepared by immersing the bare heater in the same boiling nanofluid. The comparison is shown in Fig. 10, in which water-based Al_2O_3 and TiO_2 nanofluids are used as representatives. It shows that CHF of pure water on the nanoparticle-coated surface is at least as good as that of nanofluid on the bare

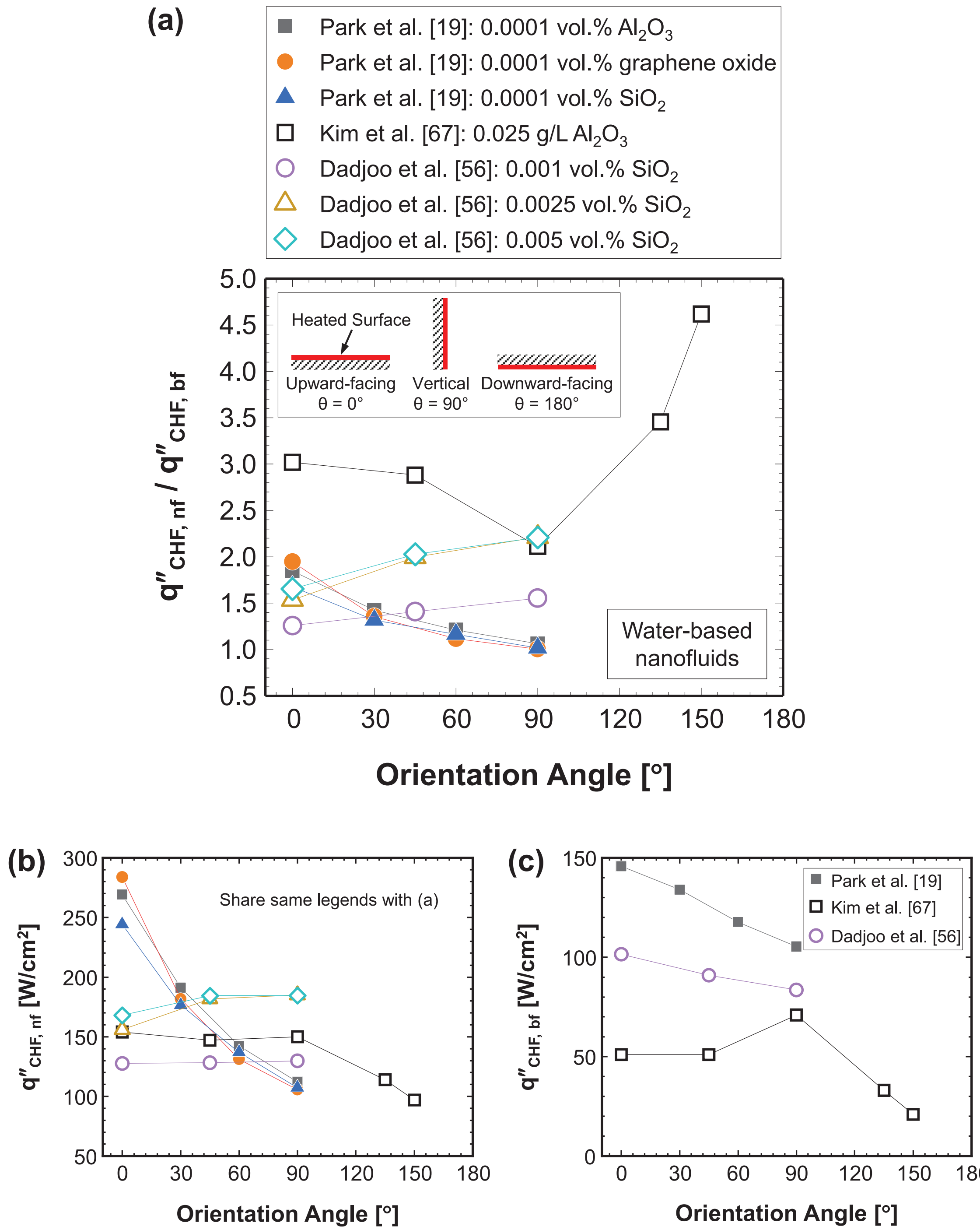

Fig. 8. Effect of heater orientation on (a) CHF ratio of nanofluid to base fluid, (b) nanofluids CHF, and (c) water CHF.

surface for all particle concentrations. This indicates that the primary reason for CHF enhancement in pool boiling of nanofluids is the changes of surface topography and microstructure due to nanoparticle coating.

Then the mechanisms behind nanoparticle deposition being responsible for enhancement of CHF are explored. A majority of researchers, including Ham et al. [57], Okawa et al. [92] Sarafraz et al. [63], and Kim and Kim [93] have found that the CHF increase of nanofluids is often accompanied by the decrease of contact angle, which therefore was used to explain the CHF enhancement as the major mechanism. To validate this mechanism, two robust methods accounting for the surface wettability are used to estimate CHF of nanofluids, including those of Kandlikar's model [94] and a correlation by Liao et al. [95],

$$q''_{CHF} = \frac{1+cos\alpha}{16}\left[\frac{2}{\pi}+\frac{\pi}{4}(1+cos\alpha)cos\theta\right]^{1/2} \times\rho_g h_{fg}\left[\sigma g\left(\rho_f-\rho_g\right)/\rho_g^2\right]^{1/4}, \tag{3a}$$

and

$$q''_{CHF} = 0.131\left[-0.73+\frac{1.73}{1+10^{-0.021\times(185.4-\theta)}}\right]\left[1+\frac{55-\alpha}{100}(0.56-0.0013\theta)\right], \times\rho_g h_{fg}\left[\sigma g\left(\rho_f-\rho_g\right)/\rho_g^2\right]^{1/4} \tag{3b}$$

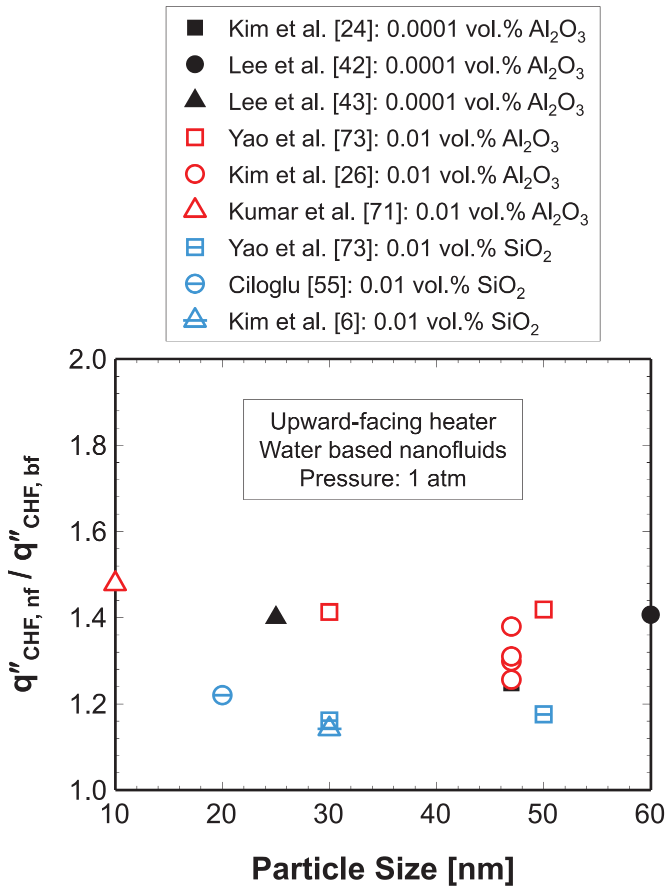

Fig. 9. Effect of nanoparticle size on CHF ratio of nanofluid to base fluid.

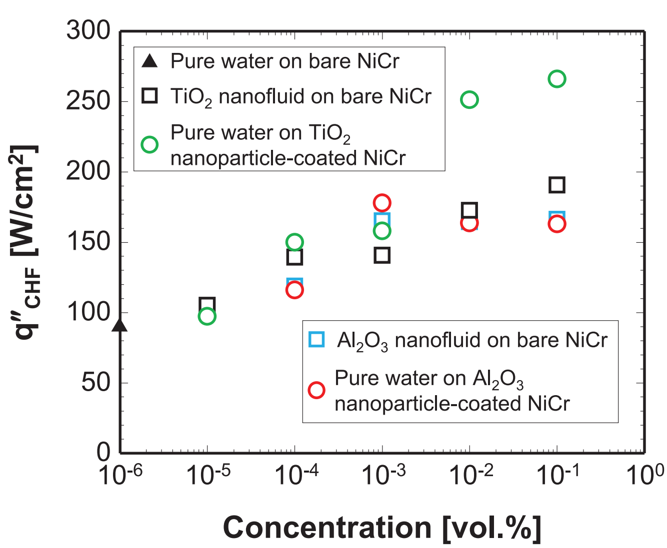

Fig. 10. Comparison of CHF enhancement of nanofluid on bare surface with pure water on nanoparticle-coated surface. Adapted from Kim et al. [91].

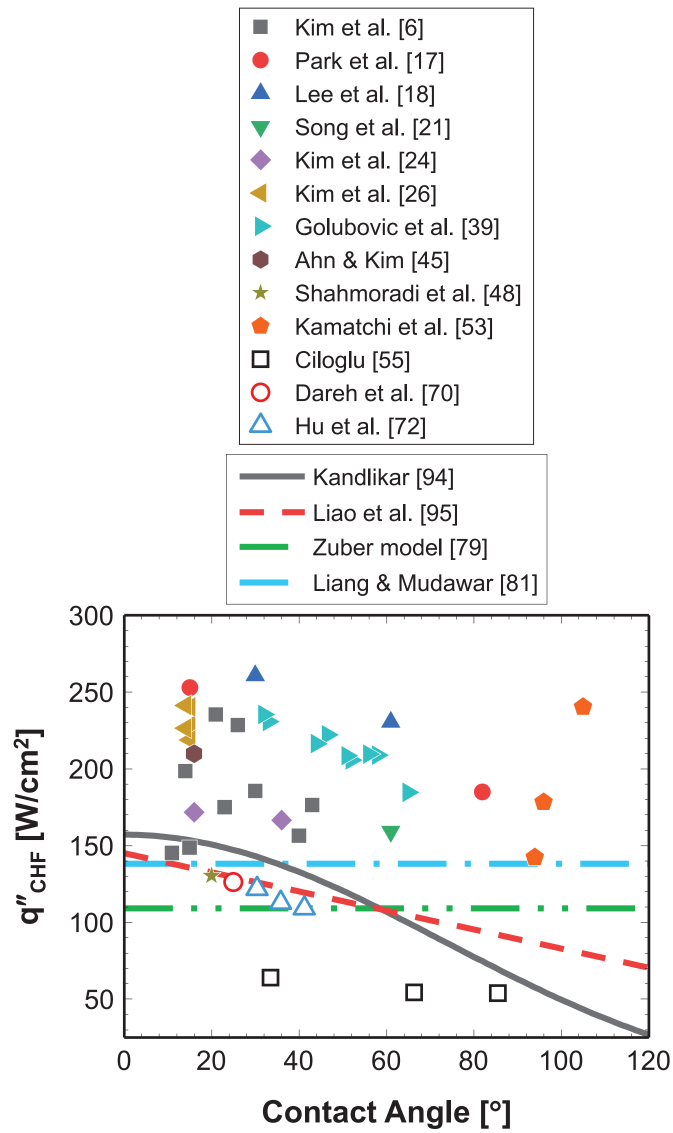

Fig. 11. Predicted CHF using Eqs. (3a) and (3b) concerning wettability effect versus measured CHF for nanofluids.

respectively, in which Kandlikar's model in Eq. (3a) is more well known. Notice that the two methods have been verified carefully by Liang and Mudawar [81,96] using another consolidated CHF database for pure liquids consisting of 800 data points amassed from 37 sources, covering contact angles from 0 to 113°. Shown in Fig. 11 is the comparison between predicted results by the two methods and experimental data extracted from Kim et al. [6], Park et al. [17], Lee et al. [18], Song et al. [21], Kim et al. [24,26], Golubovic et al. [39], Ahn and Kim [45], Shahmoradi et al. [48], Kamatchi et al. [53], Ciloglu [55], Dareh et al. [70], and Hu et al. [72]. Also included in Fig. 11 are the calculated results obtained by Eqs. (1) and (2) without consideration of wettability for reference. Fig. 11 shows that Eqs. 3(a) and (b) underestimate nanofluids pool boiling CHF apparently, mean absolute errors (MAEs) of which are 37.3% and 38.4%, respectively. This implies that the reduced contact angle or improved wettability is not the sole mechanism for the CHF enhancement.

Kim and Kim [93], and Sarafraz et al. [97] revealed that deposition of nanoparticles on the heating surface can create a porous structure, which as a consequence increases the capillary wicking. In particular to different levels of the flake layer produced by the graphene-oxide nanofluids [23], the effect of wicking will become more important. As the capillary wicking can assist liquid replenishment to the dry patches under the crowding bubble, *i.e.*, improving surface rewetting during boiling and increasing CHF, it may be another major mechanism. On the other hand, the variation of surface wettability itself is related to the capillary wicking, which makes the enhancing mechanisms become more complicated. Unfortunately, most of the works aimed for exploring the mechanisms are focused on the improved wettability induced by nanoparticle deposition on the heating surface in boiling, the capillary wicking effect is, however, less concerned. Thus, there is a need to incorporate this wicking effect into the CHF models.

Several publications, including Guan et al. [98], Kim et al. [99], Quan et al. [100], Kim et al. [101], Ahn et al. [102], and Rahman et al. [103], attempted to build up CHF models by accounting for capillary wicking effect, which were detailed in a recent review by Liang and Mudawar [104]. However, those CHF models are devised purposely for the CHF enhancement by surface modification, and many parameters, such as wicking velocity, solid or liquid fraction, porosity, size and thickness of microstructures, are difficult to be obtained. This raises a new challenge for constructing the future simple-implementation CHF models.

Besides the improved surface wettability and capillary wicking due to particle deposition, another reason being responsible for CHF enhancement might be the wettability of nanoparticles. This is experimentally validated by Quan et al. [75]. They proved that moderately hydrophilic nanoparticles not only resisted bubbles coalescence because of absorption of particles at bubble interfaces, but also roughened the surface, benefitting boiling heat transfer. In contrast, strongly hydrophilic nanoparticles did not show these positive effects. Unfortunately, the wettability of nanoparticles in most literatures is not reported. Thus, the coupling effect between improved surface wettability and nanoparticle initial wettability needs to be clarified further.

Despite a mass of available nanofluids CHF data in the literatures, the mature theoretical models and predicting methods are quite sparse. One major reason for this is lacking of good understanding of CHF enhancing mechanisms using nanofluids. Most of the explanations are qualitative, which are however not convincing. Thus, future attention should be paid to the relevant mechanisms and predicting models of nanofluids CHF, as well as the severe practical concerns prohibiting usage of nanofluids in cooling applications.

5. Conclusions

This study is dedicated to assessment of nanofluids pool boiling CHF, based on a new consolidated CHF database that is amassed from 62 sources, and consists of 431 data points covering 14 nanoparticle materials and 4 base fluids, pressures from 3 to 1100 kPa, orientation angles from 0 to 150°, and a variety of particle concentrations. Key findings from this study can be summarized as follows:

(1) For water-based nanofluids with upward-facing-orientated heater, the maximum CHF ratio of nanofluid to base fluid can achieve 4 without consideration of particles' own wettability, stabilizer, sonication, nanofluid storage time and boiling time. In contrast, the CHF ratios for base fluids of ethanol, LiBr-water solution and ethylene glycol-water solution are much lower than that of water due to their inferior thermophysical properties. However, studies using dielectric fluids as the base fluid are quite sparse.

(2) CHF is strongly dependent on the heater type (wire or block) used in boiling experiments. The direct electric-heating wire heater might be questionable due to nanoparticle deposition that distorts results of nanofluids boiling caused by electrophoresis, which migrates particles and accumulates them on the heating surface. This can be reflected by the generally higher CHF ratio using wire heater than that using block heater.

(3) The CHF ratio is reduced with increasing pressure in 3–1100 kPa, and it has a fairly weak dependence on nanoparticle size in 10–60 nm as CHF enhancement is more determined by the dispersion level of nanoparticles in the base fluid. A fine dispersion can be realized by adding surfactant, polymer or acid into the nanofluid. The effect of heater orientation angle on the CHF ratio is minor when the angle varies in 0–90°, but the ratio increases remarkably with increasing the angle in 90–180°. As very limited data is available for the orientation angle over 90°, more efforts are needed to check the increasing trend of CHF in 90–180°.

(4) CHF enhancement of nanofluids is mainly caused by the deposition of nanoparticles on the heating surface. Two major mechanisms including improved wettability and enhanced capillary wicking are responsible for the CHF enhancement. As the wicking effect is less concerned, models or correlations fail to estimate nanofluids CHF, which however warrant more studies in the future.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

Gangtao Liang: Conceptualization, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Han Yang:** Methodology, Validation, Investigation. **Jiajun Wang:** Software, Resources. **Shengqiang Shen:** Supervision, Funding acquisition.

Acknowledgements

Support of NSAF under Grant No. U2030112, National Natural Science Foundation of China under Grant No. 51876025, and Fundamental Research Funds for the Central Universities under Grant No. DUT19JC10 is gratefully acknowledged.

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